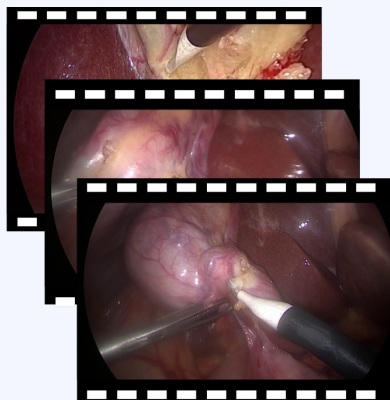




Generated Videos

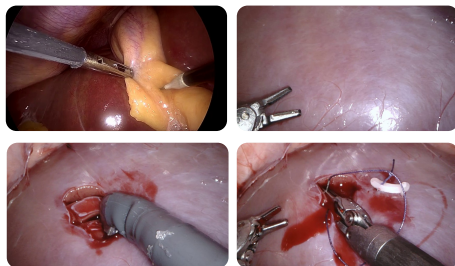


⋮

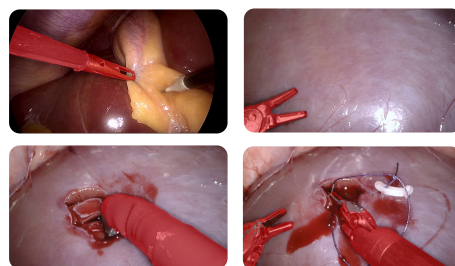
1

Data Augmentation

Generated Frames



Pseudo-labels

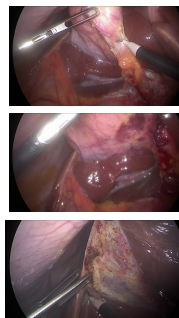


$IoU (\uparrow) 84.0$

| $Dice (\uparrow) 91.3$

2

Scene Understanding



Instrument

Action

Target



grasper

grasp

gallbladder



scissors

cut

cystic_duct



Hook

null_verb

null_verb

...

...

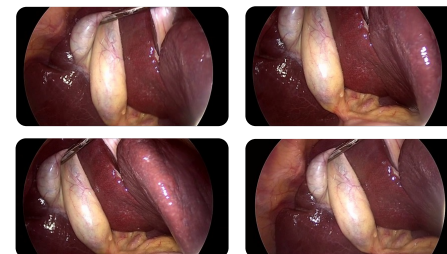
...

$AP_{IVT}(\uparrow) +6.34\%(\times 4)$

3

3D Reconstruction

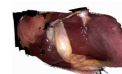
Multi-view Generated Frames



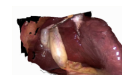
3D Reconstruction



Left



Up



Right

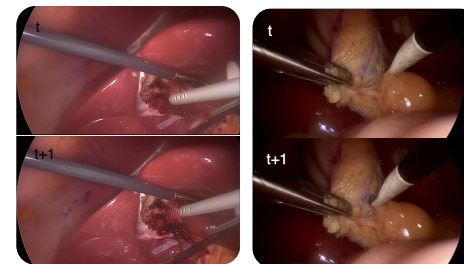


$LPIPS(\downarrow) 0.057$

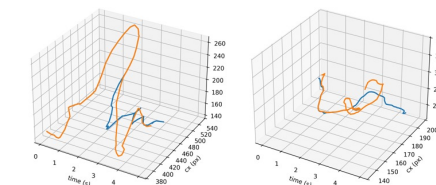
4

Trajectory Forecasting

Generated Frames



Predicated Trajectory



$RMSE(\downarrow) 14.32(1s), 19.19(5s)$